

Tutorial 11 (Additional)

1.

- (ii) The number of code error and the relative weight for each error severity class are shown in **Table 1** below:

Severity Class	Relative Weight	Number of Code Errors (NCE)
Low	2	250
Medium	5	150
High	11	75

Table 1

By referring to the figures given in the **Table 1**, compute the Total *NCE*, the *Weighted NCE (WCE)* for each severity class, the Total *WCE*, the *Code Error Density (CED)*, and the *Weighted CED (WCED)* if the KLOC is 65. (5 marks)

Error Severity Class	Calculation of NCE (Number of Code Errors)	Calculation of WCE (Number of Weighed Code Errors)	
		Relative weight	Weighted errors
Low	250	2	500
Medium	150	5	750
High	75	11	825
Total	475	-	2075

Total of NCE = 475

Total of WCE = 2075

CED

= NCE / KLOC

= 475 / 65

= 7.308

WCED

= WCE / KLOC

= 2075 / 65

= 31.923

2. Calculate the **Development Errors Removal Effectiveness (DERE)** if the **total number of development errors** is 65 and **Total Number of Failure Detected** in a software of a year is 300.

DERE

= NDE / (NDE + NYF)

= 65 / (65 + 300)

= 0.178

3. The following table shows the development milestone of a software.

Milestone	Hours of working
M1	240
M2	360
M3	160
M4	200
M5	60

Assumed that the program size is 250 KLOC, and 20% of the codes are reused from previous project. The new project has produced 580 pages of documents and 120 pages are reused from previous project. Calculate DevP, CRe, DocRe.

DevP

$$\begin{aligned}
 &= \text{DevH} / \text{KLOC} \\
 &= (240 + 360 + 160 + 200 + 60) / 250 \\
 &= 1020 / 250 \\
 &= 0.178
 \end{aligned}$$

CRe

$$\begin{aligned}
 &= \text{ReKLOC} / \text{KLOC} \\
 &= (250 \times 0.2) / 250 \\
 &= 0.2
 \end{aligned}$$

DocRe

$$\begin{aligned}
 &= \text{ReDoc} / \text{NDoc} \\
 &= 120 / 580 \\
 &= 0.2069
 \end{aligned}$$

4. An online system is expected to operate 24 hours a year (365 days). However, the total time of the system that is not able to provide some services is 144 hours. From the 144 hours, 20 hours of total time where the system is not able to perform important e-commerce activities. In addition, a total of 15 hours that the system totally failed.

Calculate *Full Availability*, *Vital Availability* and *Total Unavailability* of the system.

$$\text{NYSerH} = 24 \times 365 = 8760$$

Full Availability, FA

$$\begin{aligned}
 &= (\text{NYSerH} - \text{NYFH}) / \text{NYSerH} \\
 &= (8760 - 144) / 8760 \\
 &= 0.98356
 \end{aligned}$$

Vital Availability, VitA

$$\begin{aligned}
 &= (\text{NYSerH} - \text{NYVitFH}) / \text{NYSerH} \\
 &= (8760 - 20) / 8760 \\
 &= 0.2069
 \end{aligned}$$

Total Unavailability, TUA

$$\begin{aligned}
 &= \text{NYTFH} / \text{NYSerH} \\
 &= 15 / 8760 \\
 &= 0.0017
 \end{aligned}$$